

Programs for algebraic calculation of angular momentum coupling

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Two programs are developed for the purpose of algebraic calculation of angular momentum coupling. The first program algebraically carries out the summation of products of many Clebsch–Gordan coefficients with respect to the magnetic quantum numbers, and the second one the summation of products of many $6-j$ symbols with respect to the azimuthal quantum numbers. All are done by symbolic manipulation.

PROGRAM SUMMARY

Title of program: KENTARO/KENJIRO

Catalogue number: ACGB

Program obtainable from: CPC Program Library, Queen's University of Belfast, N. Ireland (see application form in this issue)

Computer: FACOM 780; *Installation:* Computer Center, Kyushu University, Fukuoka, Japan

Operating system: FACOM OS IV/F4 MSP

Programming language used: FORTRAN 77

Memory required to execute with typical data: 50K words

No. of bits in a word: 32

No. of lines in distributed program, including test data, etc.:
KENTARO 3179; KENJIRO 3834

Keywords: angular momentum coupling, algebraic manipulation, Clebsch–Gordan coefficients, $6-j$ symbols

Nature of physical problem

The computer programs, KENTARO and KENJIRO, allow algebraic calculation of the angular momentum coupling. The first program can execute the summation of products of many Clebsch–Gordan coefficients with respect to the magnetic quantum numbers, and the second one the summation of products of many $6-j$ symbols with respect to the azimuthal quantum numbers.

Method of solution

All calculations are performed by character operation and algebraic manipulation using various possible formulae in the angular momentum algebra.

Restrictions on the complexity of the problem

There is no special restriction, but one should note the following. Depending on the arrangement of the input data, the expression of the output results from KENTARO are not unique, although they are mathematically equivalent. To find a simpler expression, it is recommended that the results be checked by rearranging the input data. For details, refer to note 2 and 4 in section 2 of the Long Write-Up.

Typical running time

The time for the test run on the FACOM 780 is less than 0.1 s.

LONG WRITE-UP

1. Introduction

In the study of atomic physics or nuclear physics, we are often faced with tedious algebraic (non-numerical) calculations concerning the angular momentum algebra. In my long experience of theoretical nuclear study, I have wasted much time for this type of tedious calculations, and the same is probably true for others. The purpose of the two programs described below is to reduce this algebraic labour. Algebraic calculation of angular momentum coupling sometimes starts with the summation of products of many Clebsch–Gordan coefficients with respect to the magnetic quantum numbers contained therein, and proceeds on the summation of products of many 6- j symbols (or Racah coefficients) with respect to the azimuthal quantum numbers. This kind of algebraic (or formulaic) manipulation is quite simple in principle but very tedious in practice. The two computer programs execute this kind of tedious manipulations: the first program named “KENTARO” executes the summation of products of many Clebsch–Gordan coefficients, and the second one named “KENJIRO” executes the summation of products of many 6- j symbols.

Hereafter, the azimuthal quantum number and the magnetic quantum number of a general angular momentum are simply called j and m , respectively. Moreover, j sometimes stands for a set of many azimuthal quantum numbers like $\{l_1, l_2, j_1, j_2, \lambda_1, \lambda_2, \dots\}$, and m also does the similar like $\{m_1, m_2, \mu_1, \mu_2, \dots\}$.

The general form of the m -sum is written as

$$\sum_{\mu_1, \mu_2, \dots, \mu_i} (-1)^{\text{phase}(j)} (-1)^{\text{phase}(m)} \delta(\mu'_1, \mu''_1) \delta(\mu'_2, \mu''_2) \cdots \delta(\mu'_j, \mu''_j) \langle j_1 m_1 j'_1 m'_1 | j''_1 m''_1 \rangle \\ \times \langle j_2 m_2 j'_2 m'_2 | j''_2 m''_2 \rangle \langle j_3 m_3 j'_3 m'_3 | j''_3 m''_3 \rangle \cdots \langle j_k m_k j'_k m'_k | j''_k m''_k \rangle, \quad (1)$$

where $(-1)^{\text{phase}(j)}$ and $(-1)^{\text{phase}(m)}$ denote the phase factors with respect to j and m , respectively; for example,

$$(-1)^{\text{phase}(j)} = (-1)^{-j_1 + j_2 - j_3 + \cdots + j_l}, \quad (2a)$$

$$(-1)^{\text{phase}(m)} = (-1)^{m_1 - m_2 + m_3 - \cdots + m_m}, \quad (2b)$$

“KENTARO” can algebraically carry out the summation (1) using the formulae in appendix A.

On the other hand, the j -sum is generally written in a form

$$\sum_{\lambda_1, \lambda_2, \dots, \lambda_l} (-1)^{\text{phase}(j)} \delta(\lambda'_1, \lambda''_1) \delta(\lambda'_2, \lambda''_2) \cdots \delta(\lambda'_j, \lambda''_j) [\text{numerator}] \\ \times \left\{ \begin{matrix} j_1 & j'_1 & j''_1 \\ l_1 & l'_1 & l''_1 \end{matrix} \right\} \left\{ \begin{matrix} j_2 & j'_2 & j''_2 \\ l_2 & l'_2 & l''_2 \end{matrix} \right\} \cdots \left\{ \begin{matrix} j_k & j'_k & j''_k \\ l_k & l'_k & l''_k \end{matrix} \right\}. \quad (3)$$

The [numerator] is given in a form of product of components $\hat{j}_1, \hat{j}_2, \dots, \hat{j}_n$, where $\hat{j} = \sqrt{2j+1}$.

“KENJIRO” can algebraically carry out the summation (3) using the formulae in appendix B.

The programs are written in FORTRAN77. Whilst there are probably more suitable languages for algebraic computations, FORTRAN has the advantage of being well known, and the two programs described herein are fast even on a popular personal computer. For personal computers are recommended to customize the input/output part of the programs to be more “user-friendly”. For example change the I/O part to be in conversational form. Such customization is easy, since the source programs are written in FORTRAN. Once compiled there will be no big difference between languages which have been used in the original source programs, as far as the basic algorithm is not changed.

Table 1
The input data for KENTARO (unit = 1)

Format	Variable	Comment
*		You can enter a comment on this line. Even if you have no comment, you must put a blank line here.
*	NSUMM	The number of m 's over which summation should be done. (NSUMM > 0.)
5A10	CSUMM(i)	The i th m ($i = 1, 2, \dots, \text{NSUMM}$). The minus sign “-” is severely inhibited at the top of each CSUMM(i).
*	NPHJ	The number of j 's in phase (j) of (2a). (NPHJ ≥ 0 .)
5A10	CPHJ(i)	The i th j in phase (j) ($i = 1, 2, \dots, \text{NPHJ}$). If NPHJ = 0, neglect this.
*	NPHM	The number of m 's in phase (m) of (2b). (NPHM ≥ 0 .)
5A10	CPHM(i)	The i th m in phase (m) ($i = 1, 2, \dots, \text{NPHM}$). If NPHM = 0, neglect this.
*	NDELM	The number of Kronecker's deltas involved in the summation. (NDELM ≥ 0 .) As is seen in (1), these Kronecker's deltas must be with respect to m 's (not j 's).
2A10	CDELM(i, k)	The arguments of the i th delta ($i = 1, 2, \dots, \text{NDELM}$). $k = 1$ and 2 denote the 1st and 2nd arguments of the delta, respectively. If NDELM = 0, neglect his.
*	NCG	The number of Clebsch–Gordan coefficients involved in the summation. (NCG > 0.)
6A10	CCG(i, k, l)	The arguments of the i th Clebsch–Gordan coefficient ($i = 1, 2, \dots, \text{NCG}$). $k = 1, 2$ and 3 denote the 1st, 2nd and 3rd (j, m). $l = 1$ denotes the j -component and $l = 2$ the m -component. Accordingly, when thinking of, for example, $\langle j_1 m_1 j'_1 m'_1 j''_1 m''_1 \rangle$, ($k = 1, l = 2$) denotes m_1 and ($k = 3, l = 1$) denotes j''_1 .

In the following sections, it is explained in detail how to use the programs “KENTARO” and “KENJIRO”.

2. How to use “KENTARO”

2.1. Input data

The values of the input data for “KENTARO” must be “integer” or “character”. The specification of the input data is given in table 1. The quantities starting with “N” must be positive integers or zeros, and those starting with “C” must be characters, each of which must be composed of less than 7 letters, whose value must not start with Z or + or #. All these characters are treated as mere symbols in the programs. However, a single 0 is exceptional, because eq. (A.7) in appendix A is applied for this special value.

One may understand how to make the input data set by referring to table 1, and example 1 and 2. The input data are read by unit no. 1. In preparing the input data set for “KENTARO”, the following should be noted.

Note 1: One must input “all” the m 's over which summation is carried out. What does “all” mean? Let us explain this using an example. Consider the orthonormality

$$\sum_{m_1, m_2} \langle j_1 m_1 j_2 m_2 | jm \rangle \langle j_1 m_1 j_2 m_2 | j' m' \rangle = \delta(j, j') \delta(m, m'). \quad (4)$$

In this case the summation symbol $\sum_{m_1, m_2}$ on the left-hand side can be written as $\sum_{m_1}$. These two are mathematically equivalent. However, in “KENTARO”, the latter is not allowed. One must input both m_1 and m_2 without omitting any one of them. This is what “all” means; we must input all m 's relating to the summation.

Note 2: An arbitrary order of CSUMM(i) or CCG(i, j, k) with respect to i is, in principle, permitted in the input data. However, it is recommended to range them in a natural (or easily-understandable) order as angular momentum coupling. Doing so, we have a simpler answer in general. Also refer to note 4.

Note 3: When one wants to input, for example, $(-1)^{2j}$ in phase (j), one must separate it as $(-1)^j(-1)^j$. Also do so in phase (m).

2.2. How to read the output data

The answer to the summation (1) is given in a form

$$\begin{aligned} & \sum_{\lambda_1, \lambda_2, \dots, \lambda_i} \delta(\lambda'_1, \lambda''_1) \delta(\lambda'_2, \lambda''_2) \cdots \delta(\lambda'_j, \lambda''_j) (-1)^{\text{phase}(j)} \frac{[\text{numerator}]}{[\text{denominator}]} \\ & \times \left\{ \begin{matrix} l_1 & l'_1 & l''_1 \\ k_1 & k'_1 & k''_1 \end{matrix} \right\} \left\{ \begin{matrix} l_2 & l'_2 & l''_2 \\ k_2 & k'_2 & k''_2 \end{matrix} \right\} \cdots \left\{ \begin{matrix} l_k & l'_k & l''_k \\ k_k & k'_k & k''_k \end{matrix} \right\} \\ & \times \sum_{\mu_1, \mu_2, \dots, \mu_i} (-1)^{\text{phase}(m)} \delta(\mu'_1, \mu''_1) \delta(\mu'_2, \mu''_2) \cdots \delta(\mu'_m, \mu''_m) \\ & \times \langle j_1 m_1 j'_1 m'_1 | j''_1 m''_1 \rangle \langle j_2 m_2 j'_2 m'_2 | j''_2 m''_2 \rangle \cdots \langle j_n m_n j'_n m'_n | j''_n m''_n \rangle. \end{aligned} \quad (5)$$

[numerator] and [denominator] are given in a form of product of elements $\hat{j}_1, \hat{j}_2, \dots, \hat{j}_n$, which are represented as $\#j_1, \#j_2, \dots, \#j_n$ in the output; i.e.

$$\#j_1 = \hat{j}_1 = \sqrt{2j_1 + 1}, \dots, \#j_n = \hat{j}_n = \sqrt{2j_n + 1}.$$

In the output, there sometimes appear newly created j and m , which are expressed in the forms ZJ n and ZM n ($n = 1, 2, \dots$). The azimuthal and magnetic quantum numbers with the same n correspond to each other. As for the other quantities, one may easily understand what each of them means.

The output data are written on the display in a TSS session (or on the output paper in a batch job) in the same order as that in eq. (5). If you allocate a data set on unit no. 2, then the output data are simultaneously written on the data set, which can be used as a input data set for “KENJIRO”.

Note 4: The expression of the answer (5) given by “KENTARO” is not unique in general. We may have various types of expressions of the answer. Some of them are simpler and the others are not. In order to look for simpler one, users of “KENTARO” are recommended to try various ways of input by changing the order of CSUMM(i).

Note 5: An unlucky case may exceptionally happen where the summation cannot be carried out depending on the order of summations over m 's. When you encounter such an unlucky case, retry “KENTARO” after changing the order of CSUMM(i).

2.3. Example 1

Now we intend to carry out the summation on the left-hand side of the following equation using “KENTARO”:

$$\begin{aligned} & \sum_{\mu_1, \mu_2, \mu_3, m_1, m_3, m'_3, m, m_5} (-1)^{l_2 \mu_2} \delta(m_3, m'_3) \delta(m_4, -m_5) \langle j_1 m_1 l_1 \mu_1 | l_2 \mu_2 \rangle \langle l_2 \mu_2 j_4 m_4 | j m \rangle \\ & \times \langle l_3 \mu_3 j_5 m_5 | j - m \rangle \langle l_1 \mu_1 l_3 \mu_3 | j_3 m_3 \rangle \langle j_1 m_1 j_3 m'_3 | j_2 m_2 \rangle \\ & = (-1)^{l_1 - l_2 + j_1 + j_5} \hat{l}_2 \hat{j}_3 (2J + 1) \left\{ \begin{matrix} j_1 & j_2 & j_3 \\ l_3 & l_1 & l_2 \end{matrix} \right\} \left\{ \begin{matrix} j_2 & j_4 & j_5 \\ J & l_3 & l_2 \end{matrix} \right\} \\ & \times (-1)^{-m_4} \langle j_5 - m_4 j_4 m_4 | j_2 - m_2 \rangle. \end{aligned} \quad (6)$$

EXAMPLE 1					
8					
MU1	MU2	MU3	M1	M3	
M3'	M	M5			
1					
L2					
1					
-MU2					
2					
M3	M3'				
M4	-M5				
5					
J1	M1	L1	MU1	L2	MU2
L2	MU2	J4	M4	J	M
L3	MU3	J5	M5	J	-M
L1	MU1	L3	MU3	J3	M3
J1	M1	J3	M3'	J2	M2

Fig. 1. Input data set for example 1.

The input data set for this example is shown in fig. 1. The output data of example 1 is shown in the Test Run Output.

2.4. Example 2

To carry out the summation on the left-hand side of

$$\begin{aligned}
 & \sum_{\mu_1, \mu_2, m_1, m_2, m'_1, m'_2} \langle l_1 \mu_1 l_2 \mu_2 | LM \rangle \langle j_1 m_1 j_2 m_2 | IK \rangle \langle j'_1 m'_1 j'_2 m'_2 | I'K' \rangle \langle j'_1 m'_1 l_1 \mu_1 | j_1 m_1 \rangle \\
 & \times \langle j'_2 m'_2 l_2 \mu_2 | j_2 m_2 \rangle = (-1)^{K'} \langle I' - K' IK | LM \rangle \sum_{J'} (-1)^{-l_1 - l_2 + j_1 - j_2 + j'_1 - j'_2 + I'} \hat{I} \hat{J}_1 \hat{J}_2 (2J' + 1) \\
 & \times \begin{Bmatrix} J' & I' & l_2 \\ j'_2 & j_2 & j'_1 \end{Bmatrix} \begin{Bmatrix} J' & l_1 & I \\ j_1 & j_2 & j'_1 \end{Bmatrix} \begin{Bmatrix} I' & I & L \\ l_1 & l_2 & J' \end{Bmatrix} \\
 & = (-1)^{I+I'+L} \hat{I} \hat{L} \hat{J}_1 \hat{J}_2 \begin{Bmatrix} j_1 & j'_1 & l_1 \\ j_2 & j'_2 & l_2 \\ I & I' & L \end{Bmatrix} \langle I'K'LM | IK \rangle, \quad (7)
 \end{aligned}$$

the input data set should be prepared as shown in fig. 2. The output data of example 2 is also shown in the Test Run Output. "KENTARO" gives the result of the first equality in (7). As seen in this result, there appears a j -summation of product of three $6-j$ symbols. If we use the data set output from

EXAMPLE 2					
6					
MU1	MU2	M1	M2	M1'	
M2'					
0					
0					
0					
5					
L1	MU1	L2	MU2	L	M
J1	M1	J2	M2	I	K
J1'	M1'	J2'	M2'	I'	K'
J1'	M1'	L1	MU1	J1	M1
J2'	M2'	L2	MU2	J2	M2

Fig. 2. Input data set for example 2.

Table 2
The input data for KENJIRO (unit = 2)

Format	Variable	Comment
*		You can enter a comment on this line. Even if you have no comment, you must put a blank line here.
*	NSUMJ	The number of j 's over which summation should be done. (NSUMJ > 0.)
5A10	CSUMJ(i)	The i th j ($i = 1, 2, \dots, \text{NSUMJ}$). The minus sign “-” is severely inhibited at the top of each CSUMJ(i).
*	NPHJ	The number of j 's in phase (j). (NPHJ ≥ 0 .)
5A10*	CPHJ(i)	The i th j in phase (j) ($i = 1, 2, \dots, \text{NPHJ}$). If NPHJ = 0, neglect this.
*	NDELJ	The number of Kronecker's deltas involved in the summation. (NDELJ ≥ 0 .) As is seen in (1), these Kronecker's deltas must be with respect to j 's.
2A10	CDELJ(i, k)	The arguments of the i th delta ($i = 1, 2, \dots, \text{NDELJ}$). $k = 1$ and 2 denote the 1st and 2nd arguments of the delta, respectively. If NDELJ = 0, neglect this.
*	NNU	The number of component involved in [numerator]. (NNU ≥ 0 .)
5A10	CNU(i)	The i th component in [numerator] ($i = 1, 2, \dots, \text{NNU}$). Every CNU(i) must start with “#” like #J1, #L2,
*	N6J	The number of 6- j symbols involved in the summation. (N6J > 0.)
6A10	C6J(i, k)	The arguments of the i th 6- j symbol. ($i = 1, 2, \dots, \text{N6J}$). $k = 1, 2, \dots, 6$ denote the 1st, 2nd, ..., 6th arguments. As for the order of them, refer to (8).

“KENTARO” on unit no. 2 as a input data set for “KENJIRO”, then this summation can be carried out and the final result in (7) is obtained. (See example 4 shown later.)

3. How to use “KENJIRO”

3.1. Input data

The values of the input data for “KENJIRO” must be “integer” or “character”. The specification of the input data is given in table 2. The quantities starting with “N” must be positive integers or zeros, and those starting with “C” must be characters, each of which must be composed of less than 7 letters, whose value must not start with +, and with # except for CNU(i). All these characters are treated as mere symbols in the programs. However, a single 0 is exceptional, because eq. (B.3) in appendix B is used for this special value.

One may understand how to make the input data set by referring to table 2 and example 3. The order of the six arguments of the i th 6- j symbol is defined as follows:

$$\left\{ \begin{array}{ccc} \text{C6J}(i, 1) & \text{C6J}(i, 2) & \text{C6J}(i, 3) \\ \text{C6J}(i, 4) & \text{C6J}(i, 5) & \text{C6J}(i, 6) \end{array} \right\}. \quad (8)$$

In preparing the input data set for “KENJIRO”, note 3 in the previous section should be noticed. The input data set is read by unit no. 2.

3.2. How to read the output data

The answer to the summation (3) is given in the form

$$\sum_{\lambda_1, \lambda_2, \dots, \lambda_l} (-1)^{\text{phase}(j)} \delta(\lambda'_1, \lambda''_1) \delta(\lambda'_2, \lambda''_2) \cdots \delta(\lambda'_j, \lambda''_j) \frac{[\text{numerator}]}{[\text{denominator}]} \\ \times \left\{ \begin{array}{ccc} j_1 & j'_1 & l'_1 \\ l_1 & j''_1 & l''_1 \end{array} \right\} \left\{ \begin{array}{ccc} j_2 & j'_2 & l'_2 \\ l_2 & j''_2 & l''_2 \end{array} \right\} \cdots \left\{ \begin{array}{ccc} j_k & j'_k & l'_k \\ l_k & j''_k & l''_k \end{array} \right\} \times [\text{product of 9-}j \text{ symbols}]. \quad (9)$$

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EXAMPLE 3
1
J
2
J      J
0
2
#J      #J
3
J11      J12      J13      J23      J33      J
J23      J21      J22      J32      J12      J
J32      J33      J31      J11      J21      J

```

Fig. 3. Input data set for example 3.

The output data are written on the display in a TSS session (or on the output paper in a batch job) in the same order as that in (9).

3.3. Example 3

Now we intend to carry out the summation on the left-hand side of the following equation using “KENJIRO”:

$$\begin{aligned}
 & \sum_j (-)^{2j} (2j+1) \begin{Bmatrix} j_{11} & j_{12} & j_{13} \\ j_{23} & j_{33} & j \end{Bmatrix} \begin{Bmatrix} j_{23} & j_{21} & j_{22} \\ j_{32} & j_{12} & j \end{Bmatrix} \begin{Bmatrix} j_{32} & j_{33} & j_{31} \\ j_{11} & j_{21} & j \end{Bmatrix} \\
 &= \begin{Bmatrix} j_{11} & j_{12} & j_{13} \\ j_{21} & j_{22} & j_{23} \\ j_{31} & j_{32} & j_{33} \end{Bmatrix}. \tag{10}
 \end{aligned}$$

The input data set for this example is shown in fig. 3. The output data of example 3 is shown in the Test Run Output.

3.4. Example 4

Now, we intend to execute the j -summation appearing after the first equality sign in (7). To do this, we can use the output data from “KENTARO” on unit no. 2 as the input data for “KENJIRO”. Then, the resultant output becomes as shown in the Test Run Output. Thus we can easily get the final result on the right-hand side of (7).

Appendix A. Basic formulae used in “KENTARO”

Definition of 3- j symbol:

$$\begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} = (-)^{j_1-j_2-m_3} \hat{j}_3^{-1} \langle j_1 m_1 j_2 m_2 | j_3 - m_3 \rangle, \tag{A.1}$$

$$\langle j_1 m_1 j_2 m_2 | j_3 m_3 \rangle = (-)^{j_1-j_2+m_3} \hat{j}_3 \begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & -m_3 \end{pmatrix}, \tag{A.2}$$

$$\hat{j} = \sqrt{2j+1}.$$

Symmetry properties of 3- j symbol:

$$\begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} = \epsilon \begin{pmatrix} j_i & j_k & j_l \\ m_i & m_k & m_l \end{pmatrix}, \quad (\text{A.3})$$

$$\epsilon = \begin{cases} 1, & (i, k, l) \text{ is even permutation of } (1, 2, 3), \\ (-)^{j_i+j_k+j_l}, & (i, k, l) \text{ is odd permutation of } (1, 2, 3), \end{cases} \quad (\text{A.4})$$

$$\begin{pmatrix} j_1 & j_2 & j_3 \\ -m_1 & -m_2 & -m_3 \end{pmatrix} = (-)^{j_1+j_2+j_3} \begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix}. \quad (\text{A.5})$$

Orthogonality of 3- j symbol:

$$\sum_{m_1, m_2} \begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} \begin{pmatrix} j_1 & j_2 & j'_3 \\ m_1 & m_2 & m'_3 \end{pmatrix} = \frac{1}{2j_3+1} \delta(j_3, j'_3) \delta(m_3, m'_3). \quad (\text{A.6})$$

Special value:

$$\begin{pmatrix} j_1 & j_2 & 0 \\ m_1 & m_2 & 0 \end{pmatrix} = (-)^{j_1-m_1} \hat{j}_1^{-1} \delta(j_1, j_2) \delta(m_1, -m_2). \quad (\text{A.7})$$

Special case of orthogonality:

$$\sum_{m_1} (-)^{j_1-m_1} \begin{pmatrix} j_1 & j_1 & j_3 \\ m_1 & -m_1 & m_3 \end{pmatrix} = \hat{j}_1 \delta(j_3, 0) \delta(m_3, 0). \quad (\text{A.8})$$

Definition of 6- j symbol:

$$\begin{aligned} \left\{ \begin{matrix} j_1 & j_2 & j_3 \\ l_1 & l_2 & l_3 \end{matrix} \right\} &= \sum_{m_1, m_2, m_3} \sum_{\mu_1, \mu_2, \mu_3} (-)^{l_1+l_2+l_3+\mu_1+\mu_2+\mu_3} \begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} \\ &\quad \times \begin{pmatrix} j_1 & l_2 & l_3 \\ m_1 & \mu_2 & -\mu_3 \end{pmatrix} \begin{pmatrix} l_1 & j_2 & l_3 \\ -\mu_1 & m_2 & \mu_3 \end{pmatrix} \begin{pmatrix} l_1 & l_2 & j_3 \\ \mu_1 & -\mu_2 & m_3 \end{pmatrix}. \end{aligned} \quad (\text{A.9})$$

Relations between 3- j and 6- j symbols:

$$\begin{aligned} \begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} \begin{pmatrix} j_1 & j_2 & j_3 \\ l_1 & l_2 & l_3 \end{pmatrix} \\ = \sum_{\mu_1, \mu_2, \mu_3} (-)^{l_1+l_2+l_3+\mu_1+\mu_2+\mu_3} \begin{pmatrix} l_1 & l_2 & j_3 \\ \mu_1 & -\mu_2 & m_3 \end{pmatrix} \begin{pmatrix} l_3 & l_1 & j_2 \\ \mu_3 & -\mu_1 & m_2 \end{pmatrix} \begin{pmatrix} l_2 & l_3 & j_1 \\ \mu_2 & -\mu_3 & m_1 \end{pmatrix}, \end{aligned} \quad (\text{A.10})$$

$$\begin{aligned} \begin{pmatrix} l_1 & j_2 & l_3 \\ \mu_1 & m_2 & \mu_3 \end{pmatrix} \begin{pmatrix} j_1 & l_2 & l_3 \\ m_1 & -\mu_2 & -\mu_3 \end{pmatrix} \\ = \sum_{j_3} (-)^{l_3+j_3+\mu_1+m_1} (2j_3+1) \begin{pmatrix} j_1 & j_2 & j_3 \\ l_1 & l_2 & l_3 \end{pmatrix} \begin{pmatrix} l_1 & l_2 & j_3 \\ \mu_1 & -\mu_2 & m_3 \end{pmatrix} \begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & -m_3 \end{pmatrix}, \end{aligned} \quad (\text{A.11a})$$

$$\begin{aligned}
& \begin{pmatrix} l_3 & j_1 & l_2 \\ \mu_3 & m_1 & \mu_2 \end{pmatrix} \begin{pmatrix} j_3 & l_1 & l_2 \\ m_3 & -\mu_1 & -\mu_2 \end{pmatrix} \\
&= \sum_{j_2} (-)^{l_2+j_2+\mu_3+m_3} (2j_2+1) \begin{pmatrix} j_1 & j_2 & j_3 \\ l_1 & l_2 & l_3 \end{pmatrix} \begin{pmatrix} l_3 & l_1 & j_2 \\ \mu_3 & -\mu_1 & m_2 \end{pmatrix} \begin{pmatrix} j_3 & j_1 & j_2 \\ m_3 & m_1 & -m_2 \end{pmatrix}, \quad (\text{A.11b})
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} l_2 & j_3 & l_1 \\ \mu_2 & m_3 & \mu_1 \end{pmatrix} \begin{pmatrix} j_2 & l_3 & l_1 \\ m_2 & -\mu_3 & -\mu_1 \end{pmatrix} \\
&= \sum_{j_1} (-)^{l_1+j_1+\mu_2+m_2} (2j_1+1) \begin{pmatrix} j_1 & j_2 & j_3 \\ l_1 & l_2 & l_3 \end{pmatrix} \begin{pmatrix} l_2 & l_3 & j_1 \\ \mu_2 & -\mu_3 & m_1 \end{pmatrix} \begin{pmatrix} j_2 & j_3 & j_1 \\ m_2 & m_3 & -m_1 \end{pmatrix}. \quad (\text{A.11c})
\end{aligned}$$

Appendix B. Basic formulae used in “KENJIRO”

Symmetry properties of 6- j symbol:

$$\begin{pmatrix} j_1 & j_2 & j_3 \\ l_1 & l_2 & l_3 \end{pmatrix} = \begin{pmatrix} j_i & j_k & j_m \\ l_i & l_k & l_m \end{pmatrix} = \begin{pmatrix} l_i & l_k & j_m \\ j_i & j_k & l_m \end{pmatrix}, \quad (\text{B.1})$$

where (i, k, m) is any permutation of $(1, 2, 3)$.

Orthogonality:

$$\sum_{j_3} (2j_3+1) \begin{pmatrix} j_1 & j_2 & j_3 \\ l_1 & l_2 & l_3 \end{pmatrix} \begin{pmatrix} j_1 & j_2 & j_3 \\ l_1 & l_2 & l'_3 \end{pmatrix} = \frac{\delta(l_3, l'_3)}{2l_3+1}. \quad (\text{B.2})$$

Special value:

$$\begin{pmatrix} j_1 & j'_1 & 0 \\ j_2 & j'_2 & j_3 \end{pmatrix} = (-)^{j_1+j_2+j_3} (\hat{j}_1 \hat{j}_2)^{-1} \delta(j_1, j'_1) \delta(j_2, j'_2). \quad (\text{B.3})$$

Sum rules:

$$\sum_j (-1)^{j_1+j_2+j} (2j+1) \begin{pmatrix} j_1 & j_1 & j' \\ j_2 & j_2 & j \end{pmatrix} = \hat{j}_1 \hat{j}_2 \delta(j', 0), \quad (\text{B.4})$$

$$\sum_j (2j+1) \begin{pmatrix} j_1 & j_2 & j \\ j_1 & j_2 & j' \end{pmatrix} = (-1)^{2(j_1+j_2)}, \quad (\text{B.5})$$

$$\sum_j (-)^{j+j'+j''} (2j+1) \begin{pmatrix} j_1 & j_2 & j' \\ j_3 & j_4 & j \end{pmatrix} \begin{pmatrix} j_1 & j_3 & j'' \\ j_2 & j_4 & j \end{pmatrix} = \begin{pmatrix} j_1 & j_2 & j' \\ j_4 & j_3 & j'' \end{pmatrix}, \quad (\text{B.6})$$

$$\begin{aligned}
& \sum_k (-)^{k_1+k_2+k} (2k+1) \begin{pmatrix} j_1 & j'_1 & k \\ j'_2 & j_2 & j \end{pmatrix} \begin{pmatrix} k_1 & k_2 & k \\ j'_1 & j_1 & j''_2 \end{pmatrix} \begin{pmatrix} k_1 & k_2 & k \\ j'_2 & j_2 & j''_1 \end{pmatrix} \\
&= (-)^{j_1+j_2+j'_1+j'_2+j''_1+j''_2+j} \begin{pmatrix} j_1 & j_2 & j \\ j'_1 & j'_2 & k_1 \end{pmatrix} \begin{pmatrix} j'_1 & j'_2 & j \\ j''_1 & j''_2 & k_2 \end{pmatrix}. \quad (\text{B.7})
\end{aligned}$$

Definition of 9- j symbol:

$$\begin{Bmatrix} j_{11} & j_{12} & j_{13} \\ j_{21} & j_{22} & j_{23} \\ j_{31} & j_{32} & j_{33} \end{Bmatrix} = \sum_j (-)^{2j} (2j+1) \begin{Bmatrix} j_{11} & j_{12} & j_{13} \\ j_{23} & j_{33} & j \end{Bmatrix} \begin{Bmatrix} j_{23} & j_{21} & j_{22} \\ j_{32} & j_{12} & j \end{Bmatrix} \begin{Bmatrix} j_{32} & j_{33} & j_{31} \\ j_{11} & j_{21} & j \end{Bmatrix}. \quad (\text{B.8})$$

Symmetry properties of 9- j symbol:

$$\begin{Bmatrix} j_{11} & j_{12} & j_{13} \\ j_{21} & j_{22} & j_{23} \\ j_{31} & j_{32} & j_{33} \end{Bmatrix} = \begin{Bmatrix} j_{11} & j_{21} & j_{31} \\ j_{12} & j_{22} & j_{32} \\ j_{13} & j_{23} & j_{33} \end{Bmatrix} = (-1)^\sigma \begin{Bmatrix} j_{1i} & j_{1j} & j_{1k} \\ j_{2i} & j_{2j} & j_{2k} \\ j_{3i} & j_{3j} & j_{3k} \end{Bmatrix} = (-1)^\sigma \begin{Bmatrix} j_{i1} & j_{i2} & j_{i3} \\ j_{j1} & j_{j2} & j_{j3} \\ j_{k1} & j_{k2} & j_{k3} \end{Bmatrix}, \quad (\text{B.9})$$

$$\sigma = \begin{cases} 0, & (i, j, k) \text{ is even permutation of } (1, 2, 3), \\ j_{11} + j_{12} + j_{13} + j_{21} + j_{22} + j_{23} + j_{31} + j_{32} + j_{33}, & \\ (i, j, k) \text{ is odd permutation of } (1, 2, 3). \end{cases} \quad (\text{B.10})$$

Orthogonality of 9- j symbols:

$$\sum_{J_{13}, J_{24}} (J_{13} + 1)(J_{24} + 1) \begin{Bmatrix} j_1 & j_2 & J_{12} \\ j_3 & j_4 & J_{34} \\ J_{13} & J_{24} & J \end{Bmatrix} \begin{Bmatrix} j_1 & j_2 & J'_{12} \\ j_3 & j_4 & J'_{34} \\ J_{13} & J_{24} & J \end{Bmatrix} = \frac{\delta(J_{12}, J'_{12})\delta(J_{34}, J'_{34})}{(2J_{12} + 1)(2J_{34} + 1)}. \quad (\text{B.11})$$

Sum rules:

$$\sum_J (2J + 1) \begin{Bmatrix} J_{12} & J_{34} & J \\ J_{13} & J_{24} & J' \end{Bmatrix} \begin{Bmatrix} j_1 & j_2 & J_{12} \\ j_3 & j_4 & J_{34} \\ J_{13} & J_{24} & J \end{Bmatrix} = (-)^{2J'} \begin{Bmatrix} j_1 & j_3 & J_{13} \\ J_{34} & J' & j_4 \end{Bmatrix} \begin{Bmatrix} j_2 & j_4 & J_{24} \\ J' & J_{12} & j_1 \end{Bmatrix}, \quad (\text{B.12})$$

$$\begin{aligned} & \sum_{J_{13}, J_{24}} (-1)^{(j_2+j_4+J_{24})+(j_2+j_3+J_{23})} (2J_{13} + 1)(2J_{24} + 1) \begin{Bmatrix} j_1 & j_2 & J_{12} \\ j_3 & j_4 & J_{34} \\ J_{13} & J_{24} & J \end{Bmatrix} \begin{Bmatrix} j_1 & j_4 & J_{14} \\ j_3 & j_2 & J_{23} \\ J_{13} & J_{24} & J \end{Bmatrix} \\ &= (-1)^{j_3+j_4+J_{34}} \begin{Bmatrix} j_1 & j_4 & J_{14} \\ j_2 & j_3 & J_{23} \\ J_{12} & J_{34} & J \end{Bmatrix}. \end{aligned} \quad (\text{B.13})$$

TEST RUN OUTPUT

Example 1

"KENTARO" (VER. 2.1)

***** INPUT DATA *****

(-1)**(L2)

SUM OVER MU1
 SUM OVER MU2
 SUM OVER MU3
 SUM OVER M1
 SUM OVER M3
 SUM OVER M3'
 SUM OVER M
 SUM OVER M5

(-1)**(-MU2)

DELTA (M3 M3')
 DELTA (M4 -M5)

C-G	(J1	M1	L1	MU1	L2	MU2	)
C-G	(L2	MU2	J4	M4	J	M	)
C-G	(L3	MU3	J5	M5	J	-M	)
C-G	(L1	MU1	L3	MU3	J3	M3	)
C-G	(J1	M1	J3	M3'	J2	M2	)

***** OUTPUT DATA *****

(-1)**(L2)
 (-1)**(-J1)
 (-1)**(-L1)
 (-1)**(-J4)
 (-1)**(-J4)
 (-1)**(L3)
 (-1)**(L3)
 (-1)**(J5)
 (-1)**(J2)
 (-1)**(J2)
 (-1)**(-J)
 (-1)**(-J)

NUMERATORS	#L2	#J	#J	#J3
------------	-----	----	----	-----

6-J	(J2	L3	L2	
	L1	J1	J3	)

6-J	(J5	J4	J2	
	L2	L3	J	)

(-1)**(-M4)

C-G	(J5	-M4	J4	M4	J2	-M2	)
-----	------	-----	----	----	----	-----	---

Example 2

"KENTARO" (VER. 2.1)

***** INPUT DATA *****

SUM OVER MU1
 SUM OVER MU2
 SUM OVER M1
 SUM OVER M2
 SUM OVER M1'
 SUM OVER M2'

C-G	(L1	MU1	L2	MU2	L	M	)
C-G	(J1	M1	J2	M2	I	K	)
C-G	(J1'	M1'	J2'	M2'	I'	K'	)
C-G	(J1'	M1'	L1	MU1	J1	M1	)
C-G	(J2'	M2'	L2	MU2	J2	M2	)

***** OUTPUT DATA *****

SUM OVER ZJ1

(-1)**(-L1)
 (-1)**(-L2)
 (-1)**(J1)
 (-1)**(-J2)
 (-1)**(J1')
 (-1)**(-J2')
 (-1)**(I')

NUMERATORS	#I	#I'	#J1	#J2	#ZJ1
	#ZJ1				

6-J	(ZJ1	I'	L2		)
	J2'	J2	J1'		

6-J	(ZJ1	L1	I		)
	J1	J2	J1'		

6-J	(I'	I	L		)
	L1	L2	ZJ1		

(-1)**(-M)
 (-1)**(K)

C-G	(I'	-K'	I	K	L	M	)
-----	------	-----	---	---	---	---	---

Example 3

"KENJIRO" (VER. 2.1)

***** INPUT DATA *****

SUM OVER J

(-1)**(J)
 (-1)**(J)

NUMERATORS	#J	#J	
6-J (J11 J23	J12 J33	J13 J	)
6-J (J23 J32	J21 J12	J22 J	)
6-J (J32 J11	J33 J21	J31 J	)

***** OUTPUT DATA *****

9-J (J11 J21 J31	J12 J22 J32	J13 J23 J33	)
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Example 4

"KENJIRO" (VER. 2.1)

***** INPUT DATA *****

SUM OVER ZJ1

(-1)**(-L1)
 (-1)**(-L2)
 (-1)**(J1)
 (-1)**(-J2)
 (-1)**(J1')
 (-1)**(-J2')
 (-1)**(I')

NUMERATORS	#I #ZJ1	#I'	#J1	#J2	#ZJ1
6-J (ZJ1 J2'	I' J2	L2 J1'			)
6-J (ZJ1 J1	L1 J2	I J1'			)
6-J (I' L1	I L2	L ZJ1			)

***** OUTPUT DATA *****

(-1)**(-L1)
 (-1)**(L2)
 (-1)**(J1)
 (-1)**(-J2)
 (-1)**(J1')
 (-1)**(-J2')
 (-1)**(-I')

NUMERATORS	#I	#I'	#J1	#J2
9-J (J2 I J1	L2 L L1	J2' I' J1'		)